Differential Eq^n (LCCDE):

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + x(t), \quad y(0) = y'(0) = 0$$
$$x(t) = e^{2t}u(t)$$

$$y(t) = y_h(t) + y_p(t)$$

$y_h(t)$: Homogeneous Sol^n $\Rightarrow x(t) = 0$

$$\Rightarrow \frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 0$$

Let $y(t) = e^{st} \Rightarrow s^2e^{st} + 3se^{st} + 2e^{st} = 0$

$$\Rightarrow \underbrace{(s^2 + 3s + 2)}_{=0} e^{st} = 0$$

$$\Rightarrow s = -1, -2$$

$$\Rightarrow y(t) = c_1e^{-t} + c_2e^{-2t}$$

$y_p(t)$: $x(t) = e^{2t}u(t)$ Let $y_p(t) = ke^{2t}, t \geq 0$

$$\Rightarrow 4ke^{2t} + 6ke^{2t} + 2ke^{2t} = 2e^{2t} + e^{2t}, t \geq 0$$

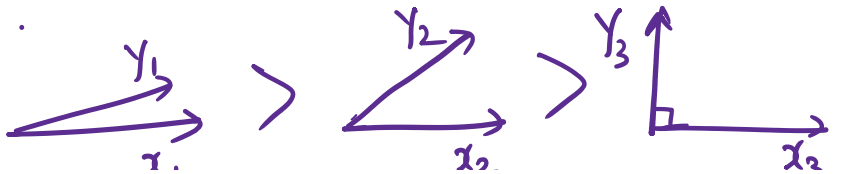
$$\Rightarrow k = 1/4 \Rightarrow y_p(t) = 1/4\, e^{2t}, t \geq 0$$

$$\Rightarrow y(t) = c_1e^{-t} + c_2e^{-2t} + \frac{1}{4}e^{2t}, t \geq 0$$

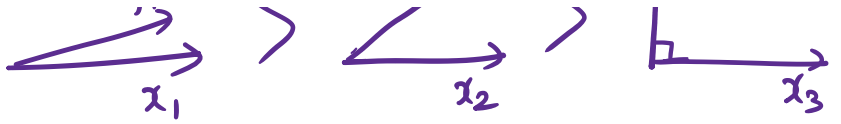
$$\left.\begin{aligned} y(0) &= 0 = c_1 + c_2 + \frac{1}{4} \\ y'(0) &= 0 = -c_1 - 2c_2 + \frac{1}{2} \end{aligned}\right\} \Rightarrow c_2 = \frac{3}{4}, c_1 = -1$$

$$\Rightarrow y(t) = -e^{-t} + \frac{3}{4}e^{-2t} + \frac{1}{4}e^{2t}, t \geq 0$$

Linear $\Rightarrow$ Initial / Auxillary Cond^n $= 0$



$$x \cdot y = x^Ty = y^Tx$$



$$\langle x,y\rangle = \frac{1}{N}\sum_{\langle N\rangle} x[n]\,y^*[n]$$

$$\langle x,y\rangle : V \times V \longrightarrow \mathbb{C}$$
$$x \quad y \longrightarrow \langle x,y\rangle$$

i) $\langle x,x\rangle \geq 0 \;\; \forall x \;\; \& = 0$ iff $x=0$

ii) $\overline{\langle x,y\rangle} = \langle y,x\rangle$

iii) $\langle \sum_i \alpha_i x_i, y\rangle = \sum_i \alpha_i \langle x_i, y\rangle$

$$\phi : S_N \longrightarrow \mathbb{C}^N$$

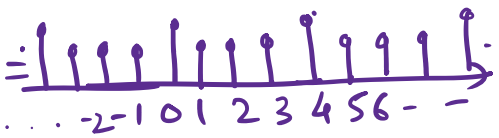
i) $\phi(x_1+x_2) = \phi(x_1) + \phi(x_2)$

ii) $\phi(\alpha x) = \alpha\,\phi(x)$

$\Big\}$ ϕ is a V.S. Homomorphism

If ϕ is one-one & onto also $\Rightarrow$ V.S. Isomorphism

$$\phi : S_N \longrightarrow \mathbb{C}^N$$

$\bar{x} :$ 

$\longrightarrow \phi(x) \triangleq \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}_{4\times 1}$

i) $\phi(x_1) = \phi(x_2) \Rightarrow x_1 = x_2 \Rightarrow$ One-one

ii) ϕ is onto

$\Big\}$ $\therefore$ <u>V.S. Isomorphism</u>